

Aidan Mulligan

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EDUCATION

University of Utah – The John and Marcia Price College of Engineering

Salt Lake City, UT

Degree: Honors Bachelor of Science in Chemical Engineering

May 2028

Emphasis: Semiconductor Processing

GPA: 3.8

Achievements: University of Utah President's Scholarship, Phillip A. and Elizabeth Dougall Endowed Scholarship, Dean's list

SKILLS/COURSEWORK

Technical: Python, Java, CAD, 3D printing, Microsoft Office Suite, Microcontrollers

Laboratory: SEM Microscopy, HPLC Purification, UV/Vis Spectroscopy, Protein Synthesis

Micromachining Course: Learned photolithography, wet/dry etching, ion implantation and diffusion, physical/chemical vapor deposition, and bulk/surface micromachining through coursework and cleanroom labs

Microsystems Design and Characterization Course: learning about the design of MEMS sensors and actuators, COMSOL simulation, and characterization techniques including SEM, ATM, & TEM

Other Notable Courses: Semiconductor Physics, Process Engineering, Thermodynamics I, Statistics, Numerical Methods

RELATED EXPERIENCE

Utah Center for Renewables, Efficiency, and Workforce (U-CREW)

Salt Lake City, UT

Research Assistant Student

January 2026-present

- Conduct industrial assessments for Utah manufacturing facilities by performing evaluations and on-site measurements
- Identify energy-reduction recommendations, proposing measures that can reduce facility energy use by $\geq 10\%$.

STARS Lab, University of Utah Department of Materials Science and Engineering

Salt Lake City, UT

Undergraduate Researcher

June 2025-September 2025

- Studied the microstructural features of high-temperature alloys for aerospace/energy applications
- Performed literature review and researched white papers to obtain grants and funding for these studies by showing how this research improves material properties for rocket engines and other applications

Barrios Lab, University of Utah Department of Biochemistry

Salt Lake City, UT

Undergraduate Researcher

October 2024-June 2025

- Worked 400 hours as a member of an interdisciplinary team focused on studying the biochemical roles of protein phosphatases in cancer and other diseases
- Performed chemical synthesis and purification to design and test a chemical sensor for SHP2 phosphatase, improving the ease of developing new anti-cancer drugs based on SHP2 phosphatase

Student Researcher

June 2023-August 2023

- Characterized unregulated antiviral drugs for feline infectious peritonitis (FIP); findings influenced FDA regulatory changes, making it easier for thousands of cats to access life-saving treatment
- Co-authored a peer-reviewed publication in the American Journal of Veterinary Research, based on this research

Metcalf Lab, University of Utah Department of Pharmacology & Toxicology

Salt Lake City, UT

Student Researcher

June 2022-August 2022

- Investigated the metabolic effects of seizures in a mouse model of a severe epilepsy (Dravet Syndrome); co-authored a peer-reviewed publication which demonstrated potential for future metabolic-based treatments

ADDITIONAL INFORMATION

University of Utah ChemE Car Club

Salt Lake City, UT

Team Member

August 2024-Present

- Collaborate with peers to design a car, competing in national and regional AIChE-sponsored ChemE Car competitions

Publications:

- Sri Hari, A., Chan, A. M., Scholl, A., **Mulligan, A.**, Camacho, J., Kearns, I. R., Opazo, G. V., Cheminant, J., Musci, T., Goh, M. J., Venosa, A., Moos, P. J., Golkowski, M., & Metcalf, C. S. (2025). Multiomic Analyses Reveal Brainstem Metabolic Changes in a Mouse Model of Dravet Syndrome. *Cells*, 15(1), 67. <https://doi.org/10.3390/cells15010067>
- **Mulligan, A. J.**, & Browning, M. E. (2024). Quality assessment and characterization of unregulated antiviral drugs for feline infectious peritonitis: implications for treatment, safety, and efficacy. *American Journal of Veterinary Research*, 85(3), 1–9. <https://doi.org/10.2460/ajvr.23.10.0221>

Other Awards:

- National Merit Scholarship, US Presidential Scholar Nominee, Valedictorian, Eagle Scout